

SecretLLM Project Report

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Abstract

This report details our development of a question-answering system for cultural-related questions based on the Meta-Llama-3-8B model for the "Behind the Secrets of Large Language Models" course. The project aims to improve model accuracy across two distinct tasks: Multiple Choice Questions (MCQ) and Short Answer Questions (SAQ). Our findings indicate that while fine-tuning provided a step-wise improvement of approximately 0.06 in the SAQ task, the most significant performance gains were achieved through Dynamic RAG. By retrieving semantically similar examples from a transformed and augmented training corpus, we achieved a final accuracy increase of 0.16 in the SAQ task compared to the baseline. Conversely, experiments with external internet research via the DuckDuckGo API negatively impacted performance, decreasing accuracy by up to 0.05 due to the introduction of non-specific noise. Finally, we discuss technical hurdles, such as data shuffling during evaluation, and how the selection of Greedy Search over complex decoding strategies ensured format consistency. Our results demonstrate that within the constraints of an 8B-parameter model, localized context injection is superior to broad web-based retrieval for nuanced cultural reasoning.

1 Introduction

The rapid advancement of Large Language Models (LLMs) has revolutionized natural language understanding and generation [6]. However, despite their impressive capabilities, these models often exhibit a significant "cultural gap". Most state-of-the-art models are predominantly trained on English-language datasets originating from Western contexts, which can lead to Western-centric biases and a lack of nuance when addressing culturally specific queries [1; 3]. Ensuring that AI systems can navigate the diverse norms, traditions, and linguistic cues of global populations is a critical

challenge for developing inclusive technology.

This report describes our approach for the "Behind the Secrets of Large Language Models" benchmark, which is designed to evaluate the cultural reasoning and question-answering capabilities of a LLM. The competition presents two primary tasks:

- **Multiple Choice Questions (MCQ):** A task where the model must interpret cultural and linguistic cues to select the correct option from four possible answers (A–D).
- **Short Answer Questions (SAQ):** A more complex task requiring the model to generate the answer directly without predefined options, where the output is compared against human-annotated ground truth.

These tasks are testing the model's ability to recognize traditions and localized knowledge.

To ensure a level playing field and focus on architectural and prompting ingenuity rather than raw parameter count, the benchmark restricts all participants to the use of the Meta-Llama-3-8B model. While smaller than its 70B counterpart, Llama-3-8B offers a robust baseline for instruction following [4], but it requires specialized techniques to achieve high performance in specialized domains like cultural QA.

In this work, we present a multi-stage approach to reduce the cultural reasoning gap. Our primary contributions include:

1. **Instruction Optimization:** An iterative prompt engineering process that moved from complex structured formats (JSON) to simplified natural language instructions to maximize the 8B model's reasoning capacity.

2. **Data Augmentation:** A transformation pipeline that converted MCQ training data into SAQ formats, effectively doubling our training resources and providing a richer retrieval corpus.
3. **Dynamic Few-Shot (Dynamic RAG):** A retrieval-augmented framework that identifies and injects semantically similar examples from the training data into the inference prompt, providing the model with localized "in-context" knowledge [2] for specific cultural queries.

Through these methods, we demonstrate that strategic context injection is particularly effective for generative tasks like SAQ, resulting in a cumulative 0.16 accuracy improvement, even when classification tasks like MCQ remain at baseline levels.

2 Methodology

Our methodology followed an iterative development cycle, beginning with baseline prompt engineering and progressing toward a sophisticated Retrieval-Augmented Generation (RAG) framework. Given the constraint of using the Llama-3-8B model, our efforts focused on maximizing the model's inherent cultural knowledge through strategic context injection and fine-tuning.

2.1 Prompt Engineering and Instruction Optimization

The initial phase of our development focused on defining a robust prompt structure. We observed that Llama-3-8B is highly sensitive to instruction complexity.

2.1.1 Format Simplification

Initial attempts to enforce a JSON-formatted output for automated parsing resulted in frequent syntax errors and "instruction drift," where the model prioritized formatting over cultural accuracy. We mitigated this by simplifying the prompt to a plain-text instruction set. By removing the JSON requirement and using clear, delimited markers for the question and the answer, we observed a more stable performance and better alignment with the cultural cues provided in the prompt.

2.1.2 Static vs. Dynamic Few-Shot Prompting

While static few-shot prompting [2] (providing a fixed set of 3 examples) offered a baseline improvement, it lacked the granularity required for four

distinct cultural contexts. This led to the development of a Dynamic Few-Shot mechanism. By selecting examples that were semantically similar to the input query, we provided the model with relevant contextual grounding tailored to the specific cultural nuances of the question.

2.2 Data Augmentation and Transformation

To address the data scarcity in the SAQ task, we implemented a data augmentation pipeline. We transformed the existing MCQ training data into SAQ-style pairs. This was achieved by stripping the multiple-choice options and using the ground-truth label to create direct Question-Answer pairs. This process doubled our effective training corpus for the SAQ task and served as the backbone for our retrieval index, ensuring the model had access to a denser knowledge base during inference. The process of this MCQ-to-SAQ transformation, ranging from prompt cleaning to the extraction of the final answer pair, is schematically illustrated in Figure 1.

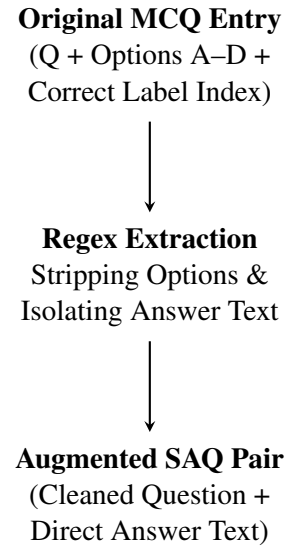


Figure 1: MCQ-to-SAQ Transformation Pipeline.

2.3 Supervised Fine-Tuning (SFT)

To align the model more closely with the specific nuances of the benchmark, we performed supervised fine-tuning on the merged training datasets. We used a low-rank adaptation (LoRA) approach [5] to maintain efficiency. The fine-tuning focused on:

- **Task Alignment:** Ensuring the model understood the distinction between selecting an option (MCQ) and generating a concise string (SAQ).

- **Cultural Nuance:** Adjusting the model’s weights to prefer culturally specific terminology found in the training set.

This phase served primarily to stabilize the model’s output format for SAQ improving the accuracy by an additional 0.06 over the initial static RAG experiments, although it did not significantly alter MCQ performance.

2.4 Dynamic Retrieval-Augmented Generation (RAG)

The most significant performance gain was achieved through a custom Dynamic RAG framework. We indexed the augmented training data using semantic embeddings. For each test query, the system retrieved the top-3 most relevant examples to be included as context in the prompt. For the SAQ task, this strategy was particularly effective. By presenting the model with similar previously answered questions, the consistency of the generated answers improved. The integration of augmented data into this retrieval loop resulted in a further accuracy boost of 0.08, leading to a total improvement of 0.16 over the baseline for the SAQ task. The architecture of our Dynamic RAG framework and the flow of information between the knowledge base and the inference process are visualized in Figure 2.

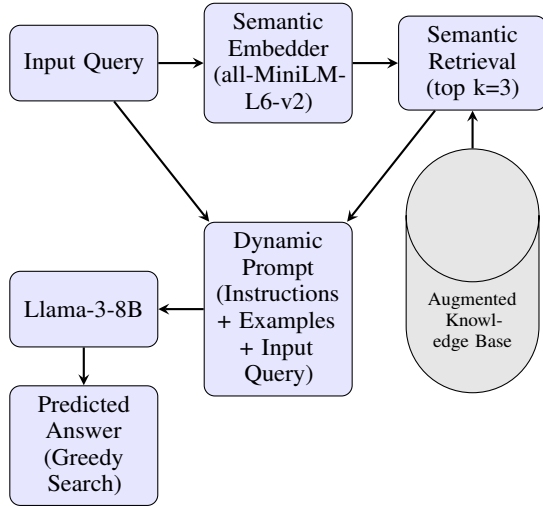


Figure 2: Dynamic RAG Architecture.

2.5 Decoding Strategies and Inference

We conducted an ablation study on decoding parameters, comparing Greedy Search, Beam Search, and Self-Consistency (majority voting over multiple paths).

- **Beam Search:** Often led to over-generalized answers that lacked the specific cultural knowledge required.
- **Self-Consistency:** While theoretically robust, it introduced significant computational overhead without a proportional gain in accuracy for this specific 8B model.

Ultimately, **Greedy Search** was selected for the final submission. It provided the most predictable outputs and adhered strictly to the concise format required by the competition’s evaluation script.

2.6 Exploration of External Knowledge

A notable experiment involved the integration of a real-time search component using the DuckDuckGoSearchAPIWrapper. We hypothesized that internet research would provide up-to-date cultural context. However, results showed a performance decrease (-0.03 to -0.05 accuracy). We concluded that the noise introduced by unfiltered search results distracted the 8B model, which struggled to synthesize conflicting information from the web compared to the high-quality, curated training data and pretrained knowledge.

3 Results and Analysis

To evaluate the impact of our interventions, we established a baseline using the out-of-the-box Llama-3-8B model with zero-shot prompting. We then sequentially applied Supervised Fine-Tuning (SFT) and our Dynamic RAG framework. The results are summarized in Table 1.

Model Configuration	MCQ Acc.	SAQ Acc.
Baseline (Zero-Shot)	0.74	0.49
Static RAG + Augmentation	0.74	0.51
Fine-Tuned (SFT)	0.74	0.57
Dynamic RAG	0.74	0.65

Table 1: Comparison of accuracy scores across MCQ and SAQ tasks.

3.1 Effectiveness of Dynamic RAG

The most notable improvement was observed in the SAQ task, where accuracy increased by 0.16, while MCQ performance remained stagnant at baseline levels. This suggests that the model’s native 8B parameters possess the necessary "knowledge," but require targeted context to align with the specific phrasing expected by human

annotators. By retrieving augmented training pairs, the model effectively learned the expected brevity and cultural context of the answers in real-time.

The stagnation in MCQ performance suggests that Llama-3-8B’s pre-trained weights are already highly optimized for discriminative tasks. In contrast, the massive gain in SAQ indicates that the model’s primary challenge was not a lack of knowledge, but an alignment issue with the specific, non-synonymous string matching used in SAQ evaluation. Our RAG framework effectively bridged this gap by providing the exact lexical references needed.

3.2 Ablation of External Search

A critical finding of our study was the negative impact of external internet search. We conducted experiments using the DuckDuckGo API to provide real-time context, assuming it would supplement the model’s internal knowledge. However, as shown in Table 2, external noise proved detrimental.

Search Configuration	Accuracy Delta
Baseline (RAG only)	0.00
Search (2 results)	-0.05
Search (25 results)	-0.03

Table 2: Impact of external search integration on model performance. All configurations resulted in a decrease in accuracy compared to the internal retrieval-only baseline.

The 8B model appeared to struggle with "distractor" information found in unfiltered search snippets. While increasing the number of search results to 25 slightly mitigated the loss (likely due to a higher chance of relevant info being present), it never outperformed the "pure" training-data RAG. This highlights that for cultural benchmarks, task-specific data is more valuable than broad web-based information.

3.3 Technical Challenges: The Shuffling Hurdle

During the development phase, we encountered a significant plateau where model accuracy remained consistently near 0.25 for the MCQ task. Upon investigation, we identified that the evaluation dataset was shuffled when loaded in. That broke the expected alignment between our local IDs and the scoring server and resulted in pure randomness.

Addressing this data alignment issue was essential. Once resolved, we were able to see the true efficacy of our fine-tuning efforts. This hurdle serves as a reminder that data integrity and pre-processing consistency are as vital as model architecture in competitive LLM benchmarks.

3.4 Decoding Strategy Analysis

Our experiments with Beam Search and Self-Consistency did not yield meaningful improvements over Greedy Search. We hypothesize that because the SAQ and MCQ tasks require high precision and adherence to a strict format, the exploratory nature of Beam Search introduced unnecessary variance that often missed the exact annotated target.

4 Conclusion

In this work, we developed and evaluated a framework for cultural question answering using the Llama-3-8B model. By navigating the constraints of a smaller parameter model, we demonstrated that the performance of such systems is not merely a function of model scale, but rather a result of strategic data alignment and context management. Our experiments highlighted several key insights:

- **Context over Scale:** The implementation of a Dynamic Few-Shot framework using augmented training data provided a substantial performance leap of +0.16 in the SAQ task, demonstrating that generative accuracy in cultural contexts is highly dependent on precise context injection and task alignment.
- **Quality over Quantity:** The failure of external internet search (−0.03 to −0.05) underscored that for nuanced cultural tasks, curated in-domain knowledge is far more effective than high-volume, noisy external data.
- **Simplicity in Prompting:** Moving away from complex JSON structures to simplified instructions proved helpful for maintaining format consistency in an 8B model.
- **Operational Integrity:** Identifying the evaluation data alignment (shuffling) issue was a critical turning point, illustrating that technical diligence in data handling is as important as algorithmic innovation.

Future work could explore more sophisticated retrieval techniques, such as hybrid dense-sparse retrieval, to further refine the relevance of the injected context. Ultimately, our results suggest that for specialized domains like cultural QA, a 'retrieval-first' strategy is especially potent for open-ended tasks where adhering to specific annotated references is critical for high evaluation scores.

References

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